

Visionary

BEHAVIOR coverage engine · follow-ups · apps · complement not replace

One-liner · NO Discord/Drive

RoboGen turns BEHAVIOR's authoring bottleneck into a queryable factory — infinite sim demos where UMI/DROID cannot afford to go — then hands hierarchical VLAs a growing skill library if verification and schema export catch up.

BEHAVIOR need → RoboGen response

Design prior for 2026 synthetic / generative-sim data

Task coverage kitchens/rooms

→ Auto task proposal on PartNet appliances + distractors

Long-horizon activities

→ Sub-task decomp + sequential chaining = proto skill library

Contact-rich articulated

→ Keep MP primitives + RL menu (Fig. 5 lesson)

Soft / deformable household

→ Traj-opt + EMD + text→3D goals

Mix with real / VLA priors

→ RoboGen demos = sim column; don't drown UMI/DROID wrists

Verification

→ Add VLM success critics before demos enter mix

Lineage to 2026 BEHAVIOR

Hand suites → OXE/DROID real → RoboGen automate curriculum

Hand suites

RLBench /
Behavior-100

taught
benchmarking

Real poles

OXE / DROID
/ UMI

taught real
diversity

RoboGen

propose→
verify→learn

taught automate
sim curriculum

2026

$\pi_{0.5}$ / N1.7
+ gen practice

schedule verify
+ gap cards

Steal the pipeline — not Genesis lock-in

propose → verify → hybrid-solve → export demo

RoboGen: Towards Unleashing Infinite Data for Automated Robot Learning via Generative Simulation

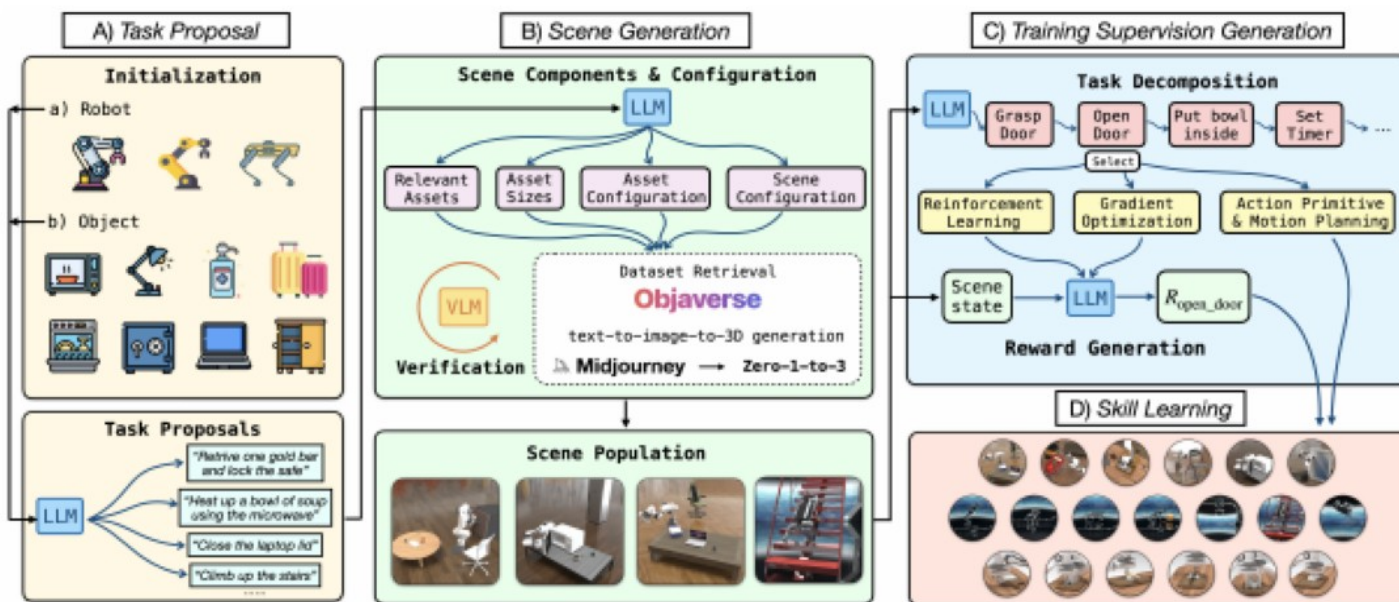


Figure 2: RoboGen consists of the following stages: A) task proposal, B) scene generation, C) training supervision generation, and D) skill learning with generated information.

from its URDF file, and 3) semantic annotations of the links in the object's articulation, e.g., which link corresponds to the door in a sampled microwave. These information are provided by the PartNetMobility dataset. Additionally, we include one example input-output pair in the prompt. We

3.2. Scene Generation

Once a task proposal is obtained, RoboGen then generates a corresponding scene for solving the task by populating the environment with a number of relevant and necessary

Motion deep dive for BEHAVIOR stacks

Planner-first, RL sparingly · export demos for VLA FT

BIT* MP

Default cheap path
approach/grasp/release
System-2 stage tool

SAC RL

Only contact/
dynamics subgoals
≤ smoke budgets

Adam/CEM

Soft + loco niches
defer if no Genesis

Export

Demos → $\pi_{0.5}$ / N1.7
NOT retrain SAC-MLP
as 2026 entry

Structure morals

Avg 3.13 substeps · ~1.5 RL + ~1.63 MP · N=8 handoff · Fig. 5: never ship RL-only on articulated BEHAVIOR stages

Follow-up research directions

1 VLM verifiers

Kill 19/155 · export schema · measurable sim2real

Scene + joint-state + skill
success critics

2 Eureka rewards

Env-feedback reward refine
inside loop

3 RLDS / Open-X export

Labeled source=robogen shard
for Octo/ π mixes

4 Skill library package

Name · lang · pre/post · algo
tag · demo blobs

5 Measurable sim2real

Held-out real appliance set +
DR upgrades

6 Open backends

Freeze versions; replace
Midjourney/GPT-4 drift

7 Mobile / multi-robot

Beyond Franka + quadruped
seeds

8 Active coverage query

Propose tasks maximizing mix
gaps

New applications

Appliance skill farms
When generative practice pays before teleop

Microwaves, dishwashers, safes
before real teleop

Soft kitchen curricula

Dough / noodles stress tests for
diff-sim + VLA FT

Quadruped loco zoos

Flip, climb, kick as synthetic
parkour pretrain

Course / challenge curricula

Emit N novel tasks overnight with
rewards

Hard-negative VLA eval

Weird articulated configs without
safety risk

Coverage gap filler

Tasks teleop cannot afford at scale

Complement — do NOT replace UMI / DROID

Real wrists stay sacred · generative = practice ground

Generative sim column

- Novel articulated / soft / loco coverage
- Failure-rich practice without hardware risk
- Stage labels + rewards attached
- Cheap to query once pipeline verified

Keep real poles

- UMI: action-diverse wild without robot logistics
- DROID: Franka scene entropy in buildings
- OXE: multi-embodiment soup gravity
- Never drown wrist demos in unfiltered synth

Mix card rule

Treat generative demos as a labeled shard beside BEHAVIOR teleop / OXE priors — Empiricist leave-synthetic-out ablation mandatory.

Gap proxy mandatory before “infinite data”

Every synthetic shard publishes DR / appearance Δ SR

H-gap

synth train $\rightarrow$ DR eval Δ SR

Verify first

size + object before SAC \$

Novel-task holdout

LLM tasks never in teleop

Refuse vanity

Self-BLEU $\neq$ useful skill

Paper's own \$5 warning

Intrinsic sim-to-real gap; large-scale automatic skill verification still hard. Visionary recipe without gap cards = rhetoric.

Anti-patterns → fixes

Explicitly **NO** Discord/Drive checklist as plan

✗ **Dump unaligned TB pick-place**

→ Use stage labels + verify + hybrid solve

✗ **Retrain SAC-MLP as 2026 entry**

→ Export demos → $\pi_{0.5}$ / GR00T N1.7

✗ **Worship endless unfiltered rollouts**

→ Coverage-as-objective + human/VLM audit

✗ **Ignore Fig. 5**

→ Planner-first on articulated stages

✗ **Skip gap honesty**

→ Publish DR Δ SR per shard

✗ **Discord/Drive as deliverable**

→ Out of scope this deck — skip

One narrative line

What 2026 should say out loud

Hand suites taught benchmarking; OXE/DROID taught real diversity; RoboGen taught automate the sim curriculum. 2026 should schedule generative practice into the data mix with verify + gap cards — not worship endless unfiltered rollouts.

Exam bite — Visionary

BEHAVIOR coverage engine · complement not replace

Steal RoboGen's propose→verify→hybrid-solve→export factory for BEHAVIOR coverage; keep MP+RL menu (Fig. 5); export demos to $\pi_{0.5}/N1.7$; publish gap proxies; complement UMI/DROID — do not replace. NO Discord/Drive. Competence = decomp + choose per sub-skill.

Numbers tattoo — pin these

Every deck · Ablation Bot MISSING (paper extract)

0.284

Self-BLEU (best)

0.165

SentBert ↓

0.193

Scene ViT ↓

0.762

Scene CLIP ↓

0.774

success / 69

19 / 155

validity fails

3.13

avg substeps

Fig. 5

RL-only collapses

0.745

manip SR (n=50)

~0.886

soft SR (n=7)

~0.833

loco SR (n=12)

~1.5+1.63

RL + MP avg

Visionary — close

Takeaways + exam bite

1. BEHAVIOR need→response: auto propose + MP/RL menu + verify.

2. Follow-ups: VLM verifiers, Eureka, RLDS export, skill library, sim2real.

3. Complement not replace UMI/DROID; gap proxy mandatory.

4. NO Discord/Drive. Export demos — don't retrain 2024 SAC-MLP.

EXAM BITE

Coverage engine beside $\pi_{0.5}/N1.7$; verify+gap; complement real poles.